

Veltrixa – Project Case Description

1. Introduction

Artificial intelligence is transforming the way organisations manage business documents by automating classification, metadata extraction, and information retrieval. While these capabilities improve efficiency and consistency, they also introduce significant privacy, security, and governance challenges.

Veltrixa Cloud is a fictional Business-to-Business (B2B) Software-as-a-Service (SaaS) platform developed as a realistic Privacy & Governance, Risk & Compliance (GRC) case study. The project demonstrates how GDPR and ISO/IEC 27001 requirements can be translated into practical governance documentation during the design and implementation of an AI-powered cloud service.

Although the organisation is fictional, the architecture, processing activities, and documentation are based on realistic enterprise practices.

2. Business Scenario

Veltrixa Cloud provides an AI-assisted document management platform for medium and large organisations operating primarily within the European Union.

Authorised users upload business documents through a secure web application. Each uploaded document is validated, scanned for malware, analysed by an internally hosted AI Classification Service, and enriched with suggested document types and metadata.

AI-generated classifications are never applied automatically. Users remain responsible for reviewing, modifying where necessary, and approving all AI-generated suggestions before they become part of downstream business processes.

This human oversight supports both business accuracy and GDPR compliance by ensuring that the AI functions as a decision-support tool rather than an autonomous decision-maker.

3. Project Objective

The objective of this project is to demonstrate how privacy, data protection, and information security documentation can be developed throughout the design and implementation of a cloud-native SaaS platform processing personal data.

The project illustrates a structured compliance approach covering:

- understanding personal data flows;
 - integrating Privacy by Design into system architecture;
 - determining whether a DPIA is required;
 - documenting processing activities;
 - establishing governance and information security controls.
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4. Solution Overview

Veltrixa Cloud consists of several integrated platform components that together support secure document processing throughout its lifecycle.

The solution includes:

- Secure Web Portal
- Upload Service
- Temporary Processing Environment
- Internal AI Classification Service
- Amazon S3 Document Storage
- PostgreSQL Metadata Repository
- Logging Platform
- Role-Based Identity Management

Uploaded documents are transmitted through encrypted HTTPS/TLS connections before entering a controlled processing environment.

The Upload Service validates each document and performs malware scanning before temporary storage within an encrypted processing bucket.

The internally hosted AI Classification Service analyses document content and generates document classification suggestions together with searchable metadata.

Original documents are stored separately from metadata to support least-privilege access principles. Metadata is maintained within PostgreSQL, while original documents are securely stored in Amazon S3.

Security monitoring, audit logging, retention management, and secure deletion are integrated throughout the platform.

5. High-Level Architecture

The solution is based on a cloud-native architecture hosted entirely within the AWS Frankfurt region.

Component	Purpose
Web Portal	User interaction and document upload
Upload Service	Validation and malware scanning
Temporary Processing	Secure short-term document processing
AI Classification Service	AI-assisted document classification and metadata generation
Amazon S3	Secure document storage

Component	Purpose
PostgreSQL	Metadata repository
Logging Platform	Audit logging and security monitoring
Identity Management	Authentication and Role-Based Access Control (RBAC)

Customer documents remain within the AWS-hosted environment throughout processing and are not used for AI model training or transferred to external AI providers.

6. Personal Data Processing

Depending on customer usage, uploaded documents may contain personal data relating to:

- Employees
- Job applicants
- Customers
- Supplier contacts
- Consultants
- Other individuals referenced within business documents
- Children (where applicable)

Typical categories of personal data include:

- Identity information
- Contact details
- Employment information
- Financial information
- Technical information
- Special category data where uploaded by customers

Veltrixa acts as Data Processor, while each customer organisation acts as Data Controller.

7. Key Design Principles

The solution has been designed according to several core privacy and security principles.

Principle	Implementation
Privacy by Design	Privacy requirements considered throughout system design

Principle	Implementation
Human Oversight	AI suggestions require user approval
Data Minimisation	Only information necessary for document management is processed
Separation of Data	Documents and metadata are stored separately
Least Privilege	Role-Based Access Control (RBAC)
Encryption	TLS in transit and AES-256 encryption at rest
EU Processing	Customer document processing remains within the EU
Secure Retention	Customer-defined retention policies and secure deletion

8. Compliance Documentation

The project documentation reflects the typical lifecycle of a privacy and information security implementation project.

Business Requirements

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Solution Design

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Data Flow Mapping

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Privacy by Design Review

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DPIA Screening

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Record of Processing Activities (ROPA)

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Information Security Management System (ISMS)

Each document builds upon the previous deliverable and demonstrates how privacy and governance activities evolve during the implementation of a new SaaS platform.

9. Project Deliverables

Document	Purpose
Data Flow Mapping	Documents how personal data flows throughout the platform.
Privacy by Design Review	Evaluates compliance with GDPR Article 25 during solution design.
DPIA Screening	Determines whether a full DPIA is required.
Record of Processing Activities (ROPA)	Documents processing activities under Article 30 GDPR.
Information Security Management System (ISMS)	Demonstrates governance aligned with ISO/IEC 27001.

10. Applicable Standards

The project has been developed with reference to:

- Regulation (EU) 2016/679 (GDPR)
 - ISO/IEC 27001:2022
 - ISO/IEC 27002:2022
 - ISO/IEC 27701
 - NIST Privacy Framework
 - Privacy by Design principles
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11. Disclaimer

Veltrixa Cloud is a fictional organisation created exclusively for educational and portfolio purposes.

The project demonstrates practical GDPR, Privacy, Governance, Risk & Compliance (GRC), and Information Security Management competencies using realistic architecture, processing activities, and industry-standard documentation